

**Created files:**

Convolution.sv

Convolution\_tb.sv

to\_greyscale.sv

to\_greyscale\_tb.sv

DE1\_SoC\_CAMERA.v (MODIFIED)

**Minilab2 Report**

In our implementation we have a greyscale module that takes 4 pixels that have been shifted out of 2 taps and into 2 registers then combines each 12 bit value and then divides by 4. We selfchecked this module with a test bench that ran for 2480 cycles to get enough data values out of the taps. Then there is a convolution module that takes three rows of pixels and multiplies each square by a sobel filter to create both a horizontal and vertical edge detection. Our test bench provided 3 values every 1280 cycles for the taps to provide actual data out of each tap and then checks that after 3840 cycles the value predicted is correct.

We ran into an issue where our camera image was being duplicated as well as what was supposed to be the left and right side of the image was overlayed. We determined that it was because our taps distance was 640 causing it to display the right side of the image to display at the same point as the left side of the image. even though the resolution ends up being 640 the image rows are still coming in at 1280.